SOLUTIONS

1. a) By using row reduction on the augmented matrix choose a, b such that the system below

$$x + ay = 2$$
$$4x + 8y = b$$

has

(i) a unique solution. **[1]**

SOLUTION

The system above has a unique solution if and only if the determinant of the system's matrix is non-zero. In that case $8 \neq 4a \Leftrightarrow a \neq 2$.

(ii) many solutions. **[1]**

SOLUTION

The system above has many solutions if the determinant of the system's matrix is zero and furthermore, the right-hand-side vector belong to the column space of the system's matrix. In that case $8 = 4a \Leftrightarrow a = 2$ and $b = 8$.

(iii) no solution. **[1]**

SOLUTION

The system above no solution if the determinant of the system's matrix is zero and furthermore, the right-hand-side vector does not belong to the column space of the system's matrix. In that case $8 = 4a \Leftrightarrow a = 2$ and $b \neq 8$.

b) Three linear systems $Ax = \underline{d}$, $Bx = \underline{e}$, $Cx = f$ have the augmented matrices

$$\begin{bmatrix} 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & 7 \\ 0 & 0 & 1 & -1 \end{bmatrix}, \begin{bmatrix} 0 & 1 & -2 & 0 & 1 \\ 0 & 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

respectively, given in reduced row echelon form.

(i) How many solutions does each system have? **[3]**

SOLUTION

The first system has a unique solution because the system's matrix is invertible.
The second system has infinitely many solutions since the rank of system's matrix is 2 and therefore, there are 2 free variables and furthermore, the right-hand-side vector belong to the column space.
The third system has no solution since the right-hand-side vector does not belong in the column space.

(ii) Which of the matrices A, B, C are invertible? **[2]**

SOLUTION

Only matrix A **is invertible.**

(iii) Find the rank of each matrix A, B, C. **[2]**

SOLUTION

From the reduced row echelon forms, we observe that the rank of A **is 3, the rank of** B **is 2 and the rank of** C **is 2.**

c) Let V be the subspace of $\Re^4$ consisting of all solutions to the system of equations written in matrix form as:

$$\begin{bmatrix} 0 & 1 & 2 & 3 \\ 3 & 2 & 1 & 0 \end{bmatrix} \begin{bmatrix} w \\ x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Let W be the orthogonal complement of V. Find row vectors $\underline{v} \in V$ and $\underline{w} \in W$ so that $\underline{v} + \underline{w} = [1 \quad 0 \quad 0 \quad 0]$. (Hint: Use the relationship $\underline{v} = [1 \quad 0 \quad 0 \quad 0]^T - \underline{w}$) **[6]**

SOLUTION

W is spanned by the rows of the system's matrix. Therefore, for some scalars s,t we have $\underline{w}=s[0\ \ 1\ \ 2\ \ 3]+t[3\ \ 2\ \ 1\ \ 0]$

We know that $\underline{v}=[1\ \ 0\ \ 0\ \ 0]-\underline{w}$ is in V. Therefore,

$$\begin{bmatrix}0&1&2&3\\3&2&1&0\end{bmatrix}\left(\begin{bmatrix}1\\0\\0\\0\end{bmatrix}-\begin{bmatrix}0&3\\1&2\\2&1\\3&0\end{bmatrix}\begin{bmatrix}s\\t\end{bmatrix}\right)=\begin{bmatrix}0\\0\end{bmatrix}$$

$$\begin{bmatrix}0&1&2&3\\3&2&1&0\end{bmatrix}\begin{bmatrix}0&3\\1&2\\2&1\\3&0\end{bmatrix}\begin{bmatrix}s\\t\end{bmatrix}=\begin{bmatrix}0&1&2&3\\3&2&1&0\end{bmatrix}\begin{bmatrix}1\\0\\0\\0\end{bmatrix}\Rightarrow\begin{bmatrix}14&4\\4&14\end{bmatrix}\begin{bmatrix}s\\t\end{bmatrix}=\begin{bmatrix}0\\3\end{bmatrix}$$

From the above system, we get

$14s+4t=0\Rightarrow 28s+8t=0$

$4s+14t=3\Rightarrow 28s+98t=21$

$$\Rightarrow t=\frac{21}{90}=\frac{7}{30}$$

$$4s+\frac{98}{30}=3\Rightarrow s=\frac{-2}{30}=\frac{-1}{15}$$

$$\underline{w}=\frac{1}{30}([0\ \ -2\ \ -4\ \ -6]+[21\ \ 14\ \ 7\ \ 0])=\frac{1}{30}[21\ \ 12\ \ 3\ \ -6]$$

$$\underline{v}=\frac{1}{30}([30\ \ 0\ \ 0\ \ 0]-[21\ \ 12\ \ 3\ \ -6])=\frac{1}{30}[9\ \ -12\ \ -3\ \ 6]$$

d) If A is a matrix of dimension 4×4, describe all vectors in the nullspace of the 4×8 matrix $B=[A\ A]$. **[4]**

SOLUTION

The nullspace of $B=[A\ A]$ contains all vectors of the form $\underline{x}=\begin{bmatrix}\underline{y}\\-\underline{y}\end{bmatrix}$ where $\underline{y}\in\Re^4$.

2. a) Consider a matrix Q which is symmetric and orthogonal. Show that the absolute value of its eigenvalues is always 1. **[2]**

SOLUTION

$Qx = \lambda x$ **and** $(Qx)^T = x^T Q^T = \lambda x^T$**. Therefore,**

$(Qx)^T Qx = \lambda^2 x^T x \Rightarrow x^T Q^T Qx = x^T Ix = x^T x = \lambda^2 x^T x \Rightarrow \lambda^2 = 1 \Rightarrow \lambda = \pm 1$

b) Consider a 3×3 matrix Q which is symmetric and orthogonal. Mark each statement (i)-(v) **True** or **False**. Justify your answer.

(i) Q does not have repeated eigenvalues. **[2]**

SOLUTION

Matrix Q has 3 eigenvalues which are 1 or -1. Therefore, it DOES have repeated eigenvalues. The statement is FALSE.

(ii) Q is always positive definite. **[2]**

SOLUTION

Matrix Q can have eigenvalues of value -1. Therefore, it cannot be always positive definite. A positive definite matrix has only positive eigenvalues. The statement is FALSE.

(iii) Q is diagonalizable. **[1]**

SOLUTION

Yes, because all symmetric and all orthogonal matrices can be diagonalized. In fact, we can choose the eigenvectors of Q to be orthogonal. The statement is TRUE.

(iv) Q is non-singular. **[1]**

SOLUTION

Yes, orthogonal matrices are all nonsingular since their column vectors are orgononal to each other and also they are square matrices. The statement is TRUE.

(v) The matrix $P = \frac{1}{2}(Q + I)$ is a projection matrix . **[2]**

SOLUTION

We can check that $P = \frac{1}{2}(Q + I)$ is symmetric and that $P^2 = P$.

$P^T = \frac{1}{2}(Q^T + I^T) = P$. Therefore, P is symmetric.

$$P^2 = \frac{1}{4}(Q + I)(Q + I) = \frac{1}{4}(QQ + 2QI + II) = \frac{1}{4}(Q^T Q + 2QI + II)$$

$$= \frac{1}{4}(I + 2Q + I) = \frac{1}{4}(2Q + 2I) = P$$

Therefore, $P = \frac{1}{2}(Q + I)$ is a projection matrix. **The statement is TRUE.**

c) Find the projection matrix onto the subspace that is formed by the vectors $[1 \quad 2 \quad 1 \quad 1]^T$ and $[-1 \quad 1 \quad 0 \quad 1]^T$. Use it to compute the projection of the vector $[1 \quad 1 \quad 2 \quad 1]^T$ onto that subspace. **[3]**

SOLUTION

Let $A=\begin{bmatrix}1 & -1\\ 2 & 1\\ 1 & 0\\ 1 & 1\end{bmatrix}$. $A^T=\begin{bmatrix}1 & 2 & 1 & 1\\ -1 & 1 & 0 & 1\end{bmatrix}$. Therefore, $A^TA=\begin{bmatrix}7 & 2\\ 2 & 3\end{bmatrix}$ and

$(A^TA)^{-1}=\frac{1}{17}\begin{bmatrix}3 & -2\\ -2 & 7\end{bmatrix}$.

We can easily show that the projection matrix is

$$A(A^TA)^{-1}A^T=\frac{1}{17}\begin{bmatrix}14 & 1 & 5 & -4\\ 1 & 11 & 4 & 7\\ 5 & 4 & 3 & 1\\ -4 & 7 & 1 & 6\end{bmatrix}.$$

Therefore, the projection of the given vector is $\frac{1}{17}\begin{bmatrix}14 & 1 & 5 & -4\\ 1 & 11 & 4 & 7\\ 5 & 4 & 3 & 1\\ -4 & 7 & 1 & 6\end{bmatrix}\begin{bmatrix}1\\ 1\\ 2\\ 1\end{bmatrix}=\frac{1}{17}\begin{bmatrix}21\\ 27\\ 16\\ 11\end{bmatrix}$.

d) Consider the matrix

$$A=\begin{bmatrix}4 & -3 & 3\\ 3 & -2 & 3\\ 3 & -3 & 4\end{bmatrix}$$

with eigenvectors

$$v_1=\begin{bmatrix}\frac{1}{\sqrt{3}}\\ \frac{1}{\sqrt{3}}\\ \frac{1}{\sqrt{3}}\end{bmatrix},\ v_2=\begin{bmatrix}\frac{1}{\sqrt{2}}\\ 0\\ -\frac{1}{\sqrt{2}}\end{bmatrix},\ v_3=\begin{bmatrix}0\\ \frac{1}{\sqrt{2}}\\ \frac{1}{\sqrt{2}}\end{bmatrix}.$$

The given order is the magnitude of the associated eigenvalue, with the first being the eigenvector that corresponds to the largest eigenvalue.

Give the diagonalized form of two distinct matrices B, C such that $B^2=A=C^2$, without calculating the inverse matrix involved in the diagonalization. **[4]**

SOLUTION

$$A-\lambda I==\begin{bmatrix}4-\lambda & -3 & 3\\ 3 & -2-\lambda & 3\\ 3 & -3 & 4-\lambda\end{bmatrix}$$

$\det(A-\lambda I)=0\Rightarrow(4-\lambda)\left[-(\lambda+2)(4-\lambda)+9\right]+3\left[3(4-\lambda)-9\right]+3\left[-9+3(\lambda+2)\right]=0\Rightarrow$

$(4-\lambda)\left[-(4\lambda-\lambda^2+8-2\lambda)+9\right]+3(12-3\lambda-9)+3(-9+3\lambda+6)=0\Rightarrow$

$(4-\lambda)(-4\lambda+\lambda^2-8+2\lambda+9)+3(3-3\lambda)+3(-3+3\lambda)=0\Rightarrow(4-\lambda)(\lambda^2-2\lambda+1)=0$

The eigenvalues of A are 4 and 1 (with multiplicity 2).

If I set $S=\begin{bmatrix}v_1 & v_2 & v_3\end{bmatrix}$, **then** $B=S\begin{bmatrix}2 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}S^{-1}$ **and** $C=S\begin{bmatrix}-2 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}S^{-1}$.

e) Suppose that the matrices A, B of dimension $m \times n$ have the same four subspaces. If they are both in reduced row echelon form, prove that F must equal G.

$$A = \begin{bmatrix} I & F \\ 0 & 0 \end{bmatrix} \quad B = \begin{bmatrix} I & G \\ 0 & 0 \end{bmatrix}$$

[3]

SOLUTION

The first row of $A = \begin{bmatrix} I & F \\ 0 & 0 \end{bmatrix}$ is a linear combination of the rows of $B = \begin{bmatrix} I & G \\ 0 & 0 \end{bmatrix}$. But since all first elements of rows of B are 0 apart from $b_{11} = 1$, it is obvious that the first row of A and B are the same. The same observation is valid for all non zero rows. Therefore, F must equal G.

3. a) Consider a matrix A of dimension 3×3 with eigenvalues $\lambda_1 = 0, \lambda_2 = c, \lambda_3 = 2$ where c is a real scalar, and associated eigenvectors

$$x_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix},\ x_2 = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix},\ x_3 = \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}$$

(i) For which values of c is matrix A diagonalizable? **[2]**

SOLUTION

The matrix is diagonalizable if it has 3 independent eigenvectors. Not only are x_1, x_2 and x_3 independent, they're orthogonal. Thus, the matrix is diagonalizable for all values of c.

(ii) For which values of c is matrix A symmetric? **[2]**

SOLUTION

We see by diagonalizing $A = S\Lambda S^T$ that we get $A = \begin{bmatrix} c+2 & -c+2 & -4 \\ -c+2 & c+2 & -4 \\ -4 & -4 & 8 \end{bmatrix}$ which is symmetric for all values of c.

(iii) For which values of c is matrix A positive definite? **[1]**

SOLUTION

All positive definite matrices are symmetric, so c must be real. The eigenvalues of a positive definite matrix must be positive. The eigenvalue 0 is not positive, so this matrix is not positive definite for any values of c.

(iv) For which values of c is matrix A positive semi-definite? **[1]**

SOLUTION

If $c = 0$ then the matrix is positive semi-definite.

(v) Is A a Markov matrix? **[2]**

SOLUTION

In a Markov matrix, one eigenvalue is 1 and the other eigenvalues are smaller than 1. Because $\lambda_3 = 2$, this cannot be a Markov matrix for any value of c.

(vi) Could $P = \frac{1}{2}A$ be a projection matrix? **[2]**

SOLUTION

The eigenvalues of $\frac{1}{2}A$ are the halves of the eigenvalues of A, therefore they are $0, \frac{c}{2}, 1$. Projection matrices are real and symmetric so their eigenvalues are real. In addition, we know that their eigenvalues are 1 and 0 because $P^2 = P$ implies $\lambda^2 = \lambda$. Thus, $\frac{1}{2}A$ could be a projection matrix if $c = 0$ or $c = 1$.

b) Consider the Singular Value Decomposition of matrix $A = U\Sigma V^T$. Suppose $\Sigma = \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}$ and U and V each have two columns.

(i) What can we say about the dimension and rand of matrix A? **[2]**

SOLUTION

We know A is a two by two matrix, and because U, Σ, V are all invertible we know that A is nonsingular and therefore, its rank is 2.

(ii) Show that if $\Sigma = \begin{bmatrix} 3 & 0 \\ 0 & 0 \end{bmatrix}$, a basis for the null space of A is the second column of V.

[2]

SOLUTION

Then A is a singular matrix of rank 1 and its null space has dimension 1. The four fundamental subspaces associated with A are spanned by orthonormal bases made up of selected columns of U, and V. In this example, the second column of V is a basis for the null space of A.

c) Consider the matrix A:

$$A = \begin{bmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{bmatrix}$$

Find the Singular Value Decomposition of matrix $A = U\Sigma V^T$. It is given that a set of orthogonal but not orthonormal eigenvectors of V is $\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 4 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ -1/2 \end{bmatrix}$. The given order corresponds to the order of the eigenvalues of V according to their magnitude with the first being the eigenvector that corresponds to the largest eigenvalue. [6]

SOLUTION

$AA^T = \begin{bmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{bmatrix} \begin{bmatrix} 3 & 2 \\ 2 & 3 \\ 2 & -2 \end{bmatrix} = \begin{bmatrix} 17 & 8 \\ 8 & 17 \end{bmatrix}$**. Its eigenvalues are obtained by the following equation:**

$(17 - \lambda^2) - 64 = 0 \Rightarrow \lambda_1 = 25, \lambda_2 = 9$ **with corresponding eigenvectors** $\begin{bmatrix} x \\ x \end{bmatrix}$ **and** $\begin{bmatrix} x \\ -x \end{bmatrix}$.

Therefore, $U = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$.

The singular values are 5 and 3. Therefore, $\Sigma = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 3 & 0 \end{bmatrix}$**. From the given eigenvectors of V we construct**

$V = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{3\sqrt{2}} & \frac{2}{3} \\ \frac{1}{\sqrt{2}} & \frac{-1}{3\sqrt{2}} & \frac{-2}{3} \\ 0 & \frac{4}{3\sqrt{2}} & \frac{1}{3} \end{bmatrix}$**. It can be verified that** $A = U\Sigma V^T$.

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